

(54) Title of the invention : MACHINE LEARNING BASED SELF ASSIST SYSTEM FOR VISUALLY IMPAIRED PERSON

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(57) Abstract :

ABSTRACT Audio output is intended for visually impaired individuals. The audio output is typically fed to an earphone, which recognizes the color and face and provides the output through the earphone. The advancement of science and innovation has been furious, and with each new development of science and innovation, the simplicity of living is expanding, but it barely addresses man's physical powerlessness. Building up support for the blind and visually impaired individuals is not a new issue. In any case, developing a computer-based solution for such a goal is establishing a zone. The computer vision zone is on its way to becoming the leader in creating vision for robots, but it is not a replacement for human vision. Despite the fact that there are various works utilizing computer vision procedures, there is currently no current technique that comprehends all of the essential prerequisites for a visually impaired individual. All of the current frameworks have been developed specifically for a specific need. In this paper, we propose another hypothetical plan that combines the significant essential capacity with some additional capacities for assisting outwardly challenged individuals. This new framework may aid in some of the critical issues confronting blind people today. Face recognition and object detection is performed in this paper for visually impaired people. Visually impaired people face numerous challenges in their daily lives. Our goal is to teach them how to live a secure and safe life for their own good. This gives them the confidence to go about their daily lives normally. Face detection enables them to recognize the faces of people they know within a certain distance. The proposed work suggests detecting and thus identifying major objects, as well as face recognition from personal datasets. The YOLO_v2 and MTCNN algorithm are used for Object and Facial Recognition, respectively. The results of image recognition are delivered to blind users via a text-to-speech library and an audio device.

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